

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details:

Batch Number : 2

Start date : 21 July 2026

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Internship Duration: 8 Weeks

1. Project Title

AI-Based Industrial Maintenance and Process Optimization System

The project is implemented as FactoryOps, a predictive maintenance and industrial monitoring platform designed to help organizations monitor machine conditions, identify possible equipment failures, manage maintenance activities, and improve operational efficiency.

2. Project Objective

The main objective of the project is to develop an AI-based industrial maintenance platform that supports proactive machine monitoring and predictive maintenance.

In traditional industrial environments, maintenance is often performed after a machine fails or according to a fixed maintenance schedule. These approaches can result in unexpected downtime, production delays, increased repair costs, and inefficient use of maintenance resources.

The proposed system, FactoryOps, aims to address these issues by providing a centralized platform for monitoring machines and analyzing their operating conditions. The system uses machine and sensor information to display equipment health, identify possible risks, and support maintenance-related decisions.

The major objectives of the project are:

- To develop a centralized dashboard for monitoring industrial machines.
- To display machine details and sensor readings in an organized manner.
- To identify machines that may be at risk of failure.
- To provide machine health and risk information.
- To support predictive maintenance activities.
- To generate machine-specific maintenance recommendations.
- To maintain and monitor operational incidents.
- To provide separate access for users and administrators.
- To connect a user-friendly frontend with a FastAPI backend.
- To improve the visibility of machine conditions and maintenance requirements.

3. Project description in detail

FactoryOps is a web-based predictive maintenance platform developed for industrial machine monitoring and process optimization. It consists of a frontend application and a backend application that communicate through APIs.

The frontend is developed using Streamlit, which provides an interactive interface for users and administrators.

The backend is developed using FastAPI, which handles API requests, machine-related data, sensor information, prediction-related operations, maintenance activities, and incident management.

The platform contains multiple functional modules.

3.1 Login and User Access

The application provides a login interface for accessing the platform. Different access roles are supported for users and administrators.

The login system helps control access to the application and allows the interface to be used according to the assigned role. After successful login, the user is directed to the main FactoryOps workspace.

3.2 Dashboard

The dashboard provides an overview of the industrial environment. It is designed to display important information in one place so that users can quickly understand the current condition of the factory. The dashboard includes information related to:

1. Overall machine condition.

2. Machine availability.
3. Operational risk.
4. Sensor activity.
5. Predicted failures.
6. Maintenance status.
7. Operational incidents.

The dashboard acts as the starting point for monitoring the factory and navigating to other modules.

3.3 Machine Monitoring

The Machines module displays information about the machines available in the industrial environment.

The module helps users view machine records and understand the current status of individual machines. It provides a structured way to monitor equipment instead of maintaining machine information manually in separate files.

Machine-related information may include:

1. Machine identification Machine name or type.
2. Current operating status.
3. Machine health information.
4. Maintenance-related details.
5. Risk or failure-related information.

This module supports better organization of the machine inventory and helps users identify machines that require attention.

3.4 Sensor Monitoring

The Sensors module is used to track sensor readings and operating conditions associated with machines.

Industrial machines generate different types of readings during operation. These readings can be useful for identifying unusual behavior or changes in machine performance.

The FactoryOps interface provides a way to view sensor information and understand machine operating conditions. Sensor information can support the identification of abnormal values and possible maintenance requirements.

Examples of industrial sensor parameters include:

1. Temperature.
2. Vibration.
3. Pressure.
4. Humidity
5. Current.
6. Other machine condition measurements.

The sensor monitoring module forms an important part of the predictive maintenance workflow because machine health and failure risk can depend on operating conditions.

3.5 Failure Prediction and Risk Analysis

The Predictions module displays predicted failures, machine health scores, and equipment risk levels.

The purpose of this module is to help users identify machines that may require inspection or preventive action. Instead of waiting for a complete machine failure, maintenance teams can use the prediction information to investigate possible problems.

The module supports the following activities:

1. Viewing predicted machine failures.
2. Checking machine health scores.
3. Understanding equipment risk levels.
4. Identifying machines that need attention.
5. Supporting preventive maintenance planning.

The prediction results should be treated as decision-support information. The maintenance team can combine prediction results with machine history, sensor readings, and operational knowledge before taking action.

3.6 Maintenance Management

The Maintenance module is used to review maintenance activities and generate machine-specific recommendations.

Maintenance recommendations can help users understand the possible action required for a machine. These recommendations may include inspection, servicing, checking sensor values, or scheduling preventive maintenance.

The module supports:

1. Reviewing maintenance records.

2. Checking the maintenance status of machines.
3. Generating machine-specific recommendations.
4. Supporting preventive maintenance planning.
5. Organizing maintenance-related information.

This functionality helps connect prediction results with practical maintenance activities.

3.7 Incident Monitoring

The Incidents module is used to monitor operational incidents and alerts.

Incidents may be associated with machine problems, abnormal sensor readings, predicted failures, or other operational issues. The module allows users to filter incidents according to their current status.

The incident monitoring functionality helps users track whether an issue is new, active, under review, or resolved. This creates a more organized process for handling machine-related problems.

3.8 Help and Backend Connection

The Help module provides information about using the FactoryOps platform. It also supports checking the connection between the frontend and the backend.

The frontend communicates with the FastAPI backend through API requests. The application includes API helper functions for retrieving data, handling responses, and managing connection-related errors.

This structure separates the user interface from the backend logic and makes the application easier to maintain and extend.

3.9 Technologies Used

The major technologies used in the project are:

1. Python: Main programming language.
2. Streamlit: Used to develop the interactive frontend.
3. FastAPI: Used to develop the backend APIs.
- 4 .REST APIs: Used for communication between the frontend and backend.
5. HTML and CSS: Used for interface customization and styling.
6. JavaScript: Used for navigation-related interface behavior.
7. Session State: Used to manage login and user navigation information.

Environment Variables: Used to configure backend URLs and application settings.

3.10 Real-World Impact

FactoryOps can be useful in manufacturing plants, production facilities, processing units, and other industrial environments where machine availability is important.

The system can support organizations by:

1. Reducing dependence on reactive maintenance.
2. Improving machine condition visibility.
3. Supporting early identification of possible failures.
4. Helping maintenance teams prioritize inspections.
5. Organizing machine and incident information.
6. Improving communication between monitoring and maintenance activities.
7. Supporting better planning of maintenance operations.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand the problem statement, predictive maintenance concepts, and project requirements.	Studied industrial maintenance challenges, machine downtime, predictive maintenance, and the expected FactoryOps features.
Week 2	Plan the system architecture and identify frontend and backend requirements.	Designed the basic structure of the FactoryOps platform and identified the modules required for machine, sensor, prediction, maintenance, and incident monitoring.
Week 3	Develop the backend and API structure.	Worked on the FastAPI backend structure and API communication required for retrieving machine and operational data.
Week 4	Develop the frontend interface and login system.	Created the Streamlit interface, login screen,

	.	role-based access flow, sidebar navigation, and main workspace structure
Week 5	Implement machine and sensor monitoring modules.	Added machine inventory display, sensor monitoring, dashboard information, and API-based data retrieval.
Week 6	Implement prediction, maintenance, and incident modules.	Added prediction display, machine health and risk information, maintenance recommendations, and incident monitoring functionality.
Week 7	Test the application and improve the user interface.	Tested frontend-backend connectivity, API responses, navigation, session handling, error handling, and visual styling.
Week 8	Finalize the project and prepare the presentation and report.	Reviewed the complete FactoryOps workflow, corrected interface issues, prepared screenshots, documented the project, and finalized the internship report.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project requirements were discussed and the industrial predictive maintenance problem was understood.	Week 1
Prototype/First Draft	The initial FactoryOps structure, frontend layout, login flow, and backend connection were prepared.	Week 4

Mid-Term Review	The main monitoring modules, including machines, sensors, and dashboard features, were reviewed.	Week 5
Final Submission	The complete application, documentation, and internship report were prepared	Week 8
Presentation	The project workflow, features, technologies, and outcomes were prepared for presentation.	Week 8

5b. Project execution details

The FactoryOps project was executed in multiple stages. The work began with understanding the industrial maintenance problem and identifying the features required in a predictive maintenance platform.

5b.1 Requirement Analysis

The first stage involved understanding the importance of machine monitoring and predictive maintenance.

The team identified the need for a system that could display machine conditions, sensor readings, predicted failures, maintenance recommendations, and incidents in one interface.

The project requirements were divided into functional modules so that each module could be developed.

5b.2 System Design

The system was designed using a frontend-backend architecture.

The Streamlit frontend is responsible for displaying information and accepting user interaction.

The FastAPI backend is responsible for providing data through APIs and handling backend operations.

The main workflow is:

- The user opens the FactoryOps application.

- The user logs in using the available credentials.
- The frontend sends API requests to the backend.
- The backend returns machine, sensor, prediction, maintenance, or incident data.
- The frontend displays the information in dashboards and tables.
- The user reviews the information and takes suitable maintenance-related action.

5b.3 Frontend Development

The frontend was developed using Streamlit. The interface was organized into separate sections so that users could access different functions easily.

The frontend includes:

- Login page.
- Dashboard.
- Machines section.
- Sensors section.
- Predictions section.
- Maintenance section.
- Incidents section.
- Help section.
- Sidebar navigation.
- Refresh data option.
- Logout option.

5b.4 Backend Integration

The FastAPI backend was connected to the Streamlit frontend using API requests.

Helper functions were used to:

- Send API requests.
- Retrieve data from backend endpoints.
- Handle API response errors.
- Process returned lists and records.
- Display useful information in the frontend.
- Manage backend connection issues.

The API integration allows the frontend to display updated information without directly handling backend operations.

5b.5 Module Integration

After developing individual modules, they were integrated into the main FactoryOps workspace.

The dashboard provides a summary of the system, while the Machines and Sensors modules provide detailed operational information. The Predictions module provides risk-related information, and the Maintenance and Incidents modules support follow-up actions.

The Help module provides basic guidance and backend connection information.

5b.6 Testing and Improvements

Testing was performed to check the functionality and usability of the application.

The following areas were reviewed:

- Login behavior.
- Role selection.
- Sidebar navigation.
- API connectivity.
- Machine data display.
- Sensor data display.
- Prediction information.
- Maintenance recommendations.
- Incident filtering.
- Refresh functionality.
- Logout functionality.
- Error handling.
- Layout and styling.

The interface was improved to make the navigation clearer and the application easier to use.

6. Snapshots / Screenshots

Screenshot 1: FactoryOps Login Page

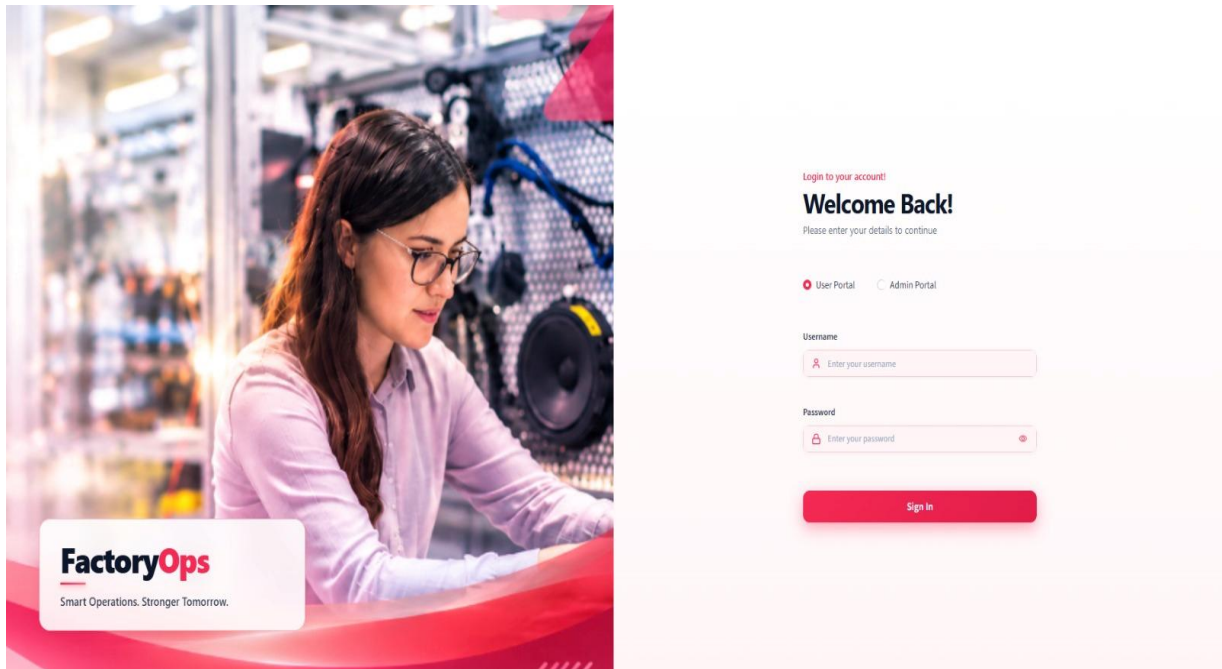


Figure 1: Login interface of the FactoryOps predictive maintenance platform.

Screenshot 2: FactoryOps Dashboard

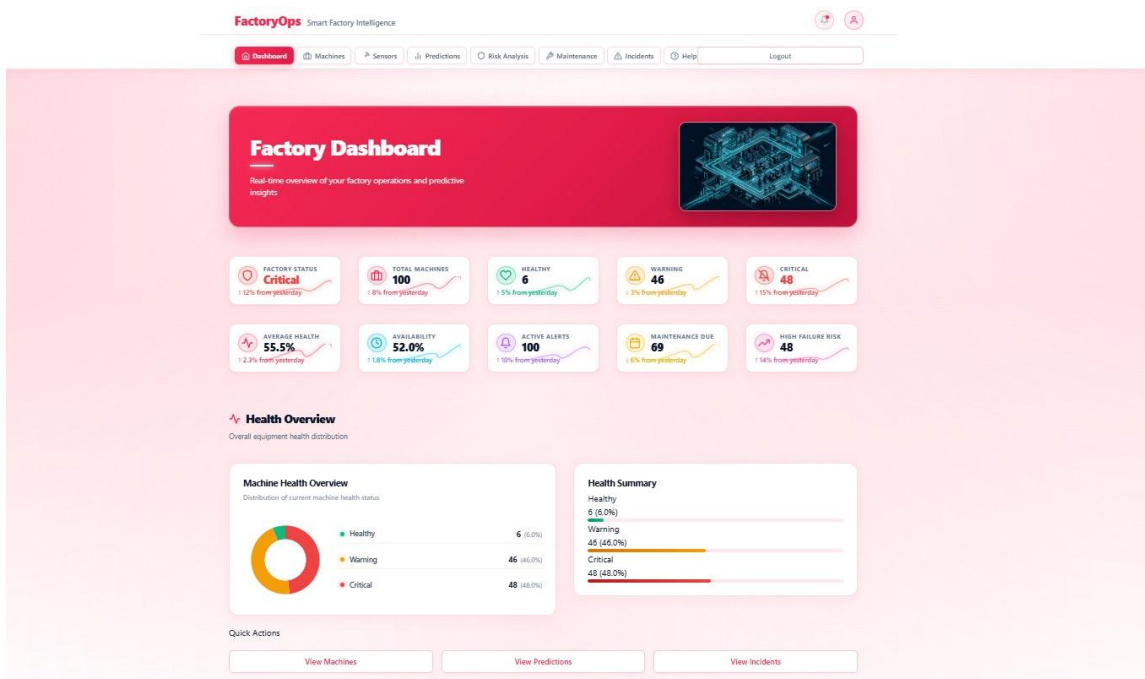


Figure 2: FactoryOps dashboard displaying the overall factory monitoring information.

Screenshot 3: Machine Monitoring Module

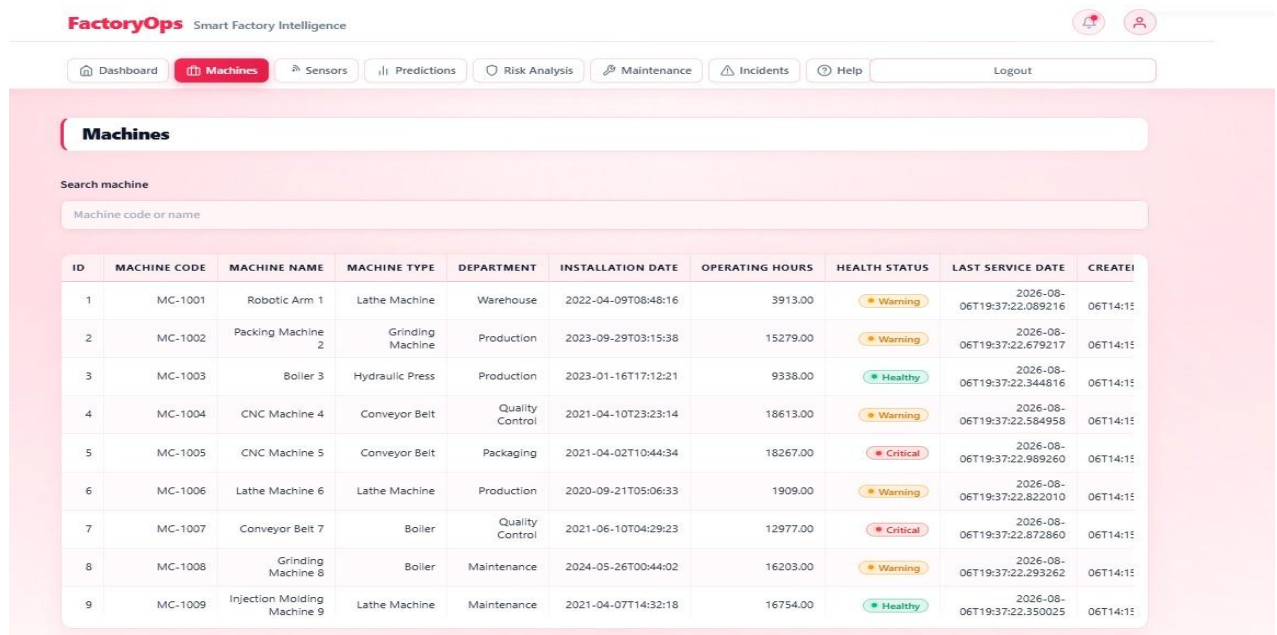


Figure 3: Machine monitoring section displaying machine records and operating status.

Screenshot 4: Sensor Monitoring Module

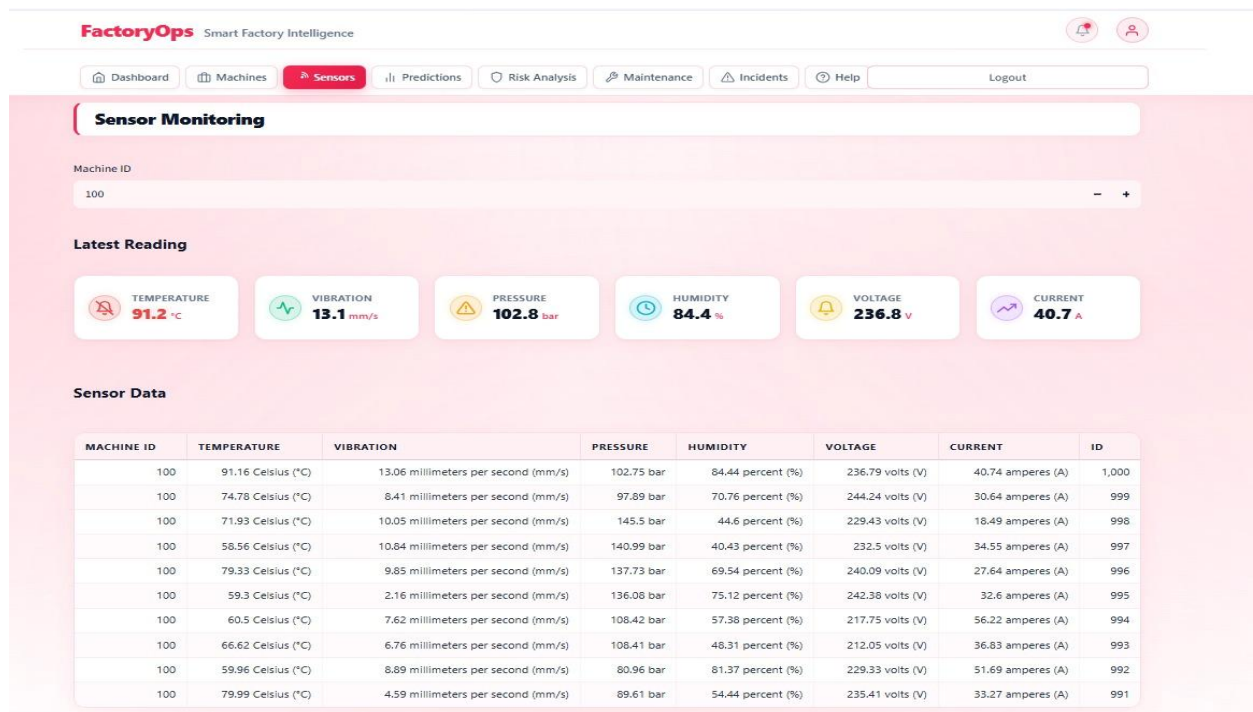


Figure 4: Sensor monitoring section displaying machine sensor readings.

Screenshot 5: Predictions Module

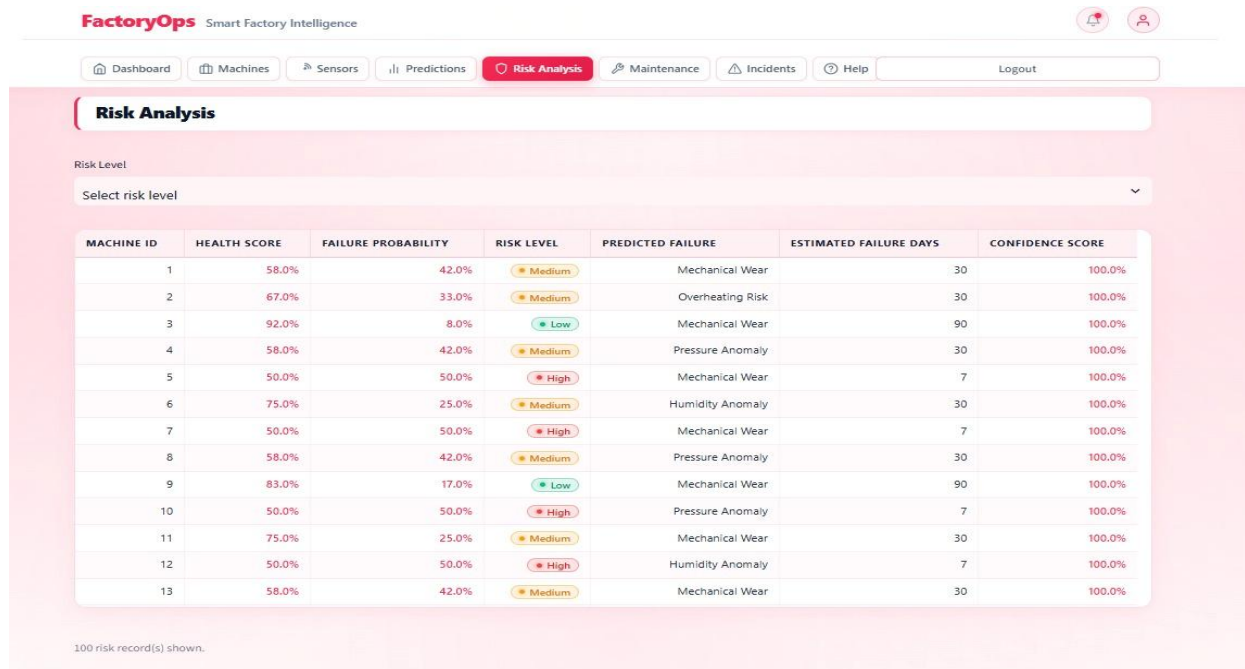


Figure 5: Prediction section displaying machine health scores and equipment risk levels.

Screenshot 6: Maintenance Module

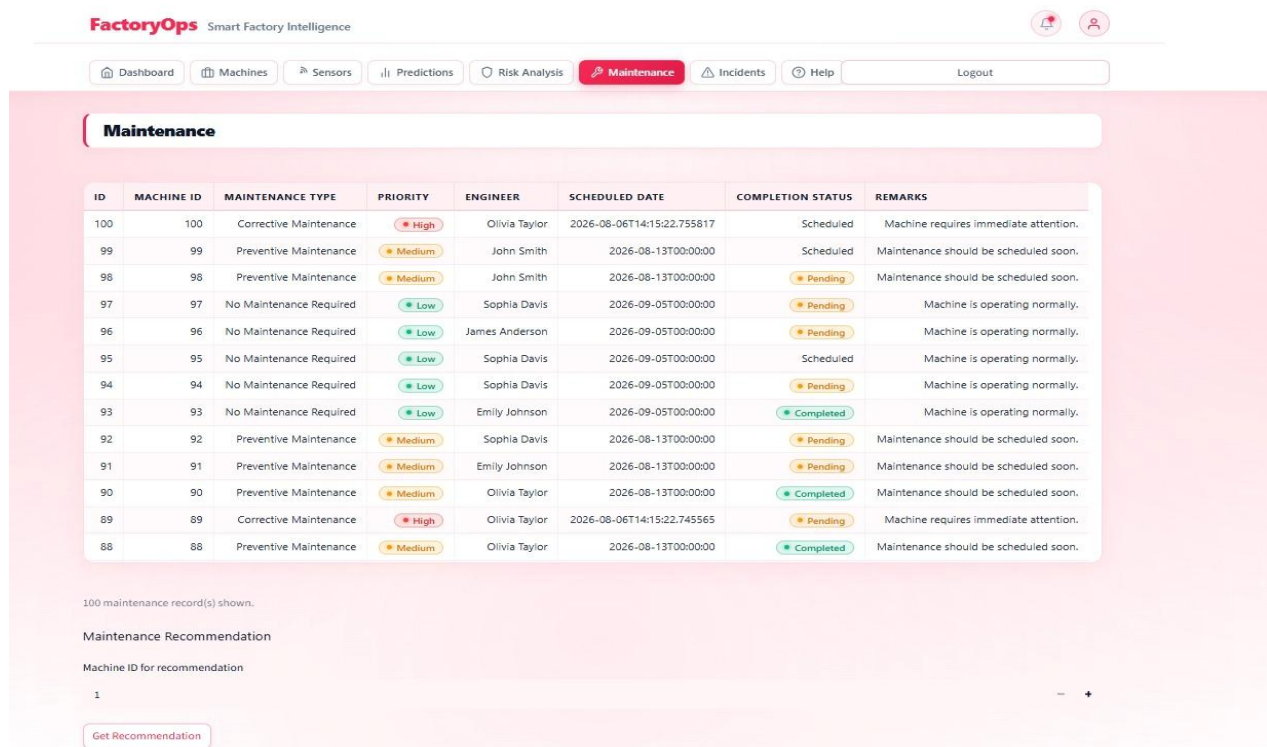


Figure 6: Maintenance section displaying maintenance activities and recommendations.

Screenshot 7: Incident Monitoring Module

FactoryOps Smart Factory Intelligence

Dashboard Machines Sensors Predictions Risk Analysis Maintenance Incidents Help Logout

Incidents & Alerts

Total Incidents
91

MACHINE ID	INCIDENT TYPE	SEVERITY	ROOT CAUSE	DESCRIPTION	ACTION TAKEN	RESOLVED STATUS
100	Machine Abnormality	Medium	Bearing Wear, Shaft Misalignment	Sensor values exceeded operating limits.	Inspect and replace machine bearings.	Under Investigation
99	Machine Abnormality	Low	No Significant Fault Detected	Sensor values exceeded operating limits.	Continue Normal Machine Operation.	Resolved
98	Machine Abnormality	Medium	Bearing Wear, Shaft Misalignment	Sensor values exceeded operating limits.	Inspect and replace machine bearings.	Resolved
97	Machine Abnormality	Medium	Shaft Misalignment	Sensor values exceeded operating limits.	Perform shaft alignment and balancing.	Pending
96	Machine Abnormality	Medium	Bearing Wear, Shaft Misalignment	Sensor values exceeded operating limits.	Inspect and replace machine bearings.	Resolved
95	Machine Abnormality	Low	Valve Blockage	Sensor values exceeded operating limits.	Inspect valves and clean pressure lines.	Pending
94	Machine Abnormality	Low	No Significant Fault Detected	Sensor values exceeded operating limits.	Continue Normal Machine Operation.	Pending
93	Machine Abnormality	Low	Valve Blockage	Sensor values exceeded operating limits.	Inspect valves and clean pressure lines.	Pending
92	Machine Abnormality	Medium	Bearing Wear, Shaft Misalignment	Sensor values exceeded operating limits.	Inspect and replace machine bearings.	Under Investigation

Figure 7: Incident monitoring section displaying operational incidents and their statuses.

Screenshot 8: Backend API or Project Structure

Predictive Maintenance API 1.0.0 OAS 3.1

API for Smart Factory Operations

default

- GET / Root
- GET /machines Get Machines
- GET /sensors Get Sensor Readings
- POST /sensors Add Sensor Reading

Predictions

- GET /predictions/ Get Predictions

Maintenance

- GET /maintenance/ Get All Maintenance

Incidents

- GET /incidents/ Get All Incidents

Figure 8: FastAPI backend structure and frontend-backend API integration.

7. Challenges Faced

During the development of FactoryOps, the team faced several technical and implementation-related challenges. Since the application involved multiple frontend modules, backend APIs, database operations, authentication, and machine prediction information, maintaining smooth communication between all components required careful testing and coordination.

7.1 Frontend-Backend Connectivity

Connecting the Streamlit frontend with the FastAPI backend required careful handling of API URLs, requests, responses, and errors.

This was addressed by creating reusable API helper functions and configuring the backend URL through an environment variable. This made it easier to manage API communication and update the backend address when required.

An additional challenge was ensuring that the frontend remained functional when the backend was temporarily unavailable or when an API request failed. Appropriate error handling was added so that users could receive meaningful messages instead of the application failing unexpectedly.

7.2 Handling API Responses

Different API endpoints may return different types of responses, such as lists, objects, or error messages. Processing these responses consistently was a challenge.

The issue was managed by using helper functions to extract lists, retrieve values safely, and handle missing or unexpected response data.

The team also had to consider cases where an API response might contain incomplete data or an empty result. Safe access methods and validation helped prevent errors while displaying information on the frontend.

7.3 Maintaining a Clear User Interface

The application contains several modules, including machines, sensors, predictions, maintenance, and incidents.

Displaying all these modules in a clear manner required careful layout planning.

The team organized the application into separate sections and added sidebar navigation to improve usability.

7.4 Session and Login Management

Maintaining login status and user navigation across different sections required session-state handling.

This was managed using Streamlit session state variables for authentication, username, role, and the selected page.

The team also had to ensure that the login state was maintained while navigating between different sections. Logout functionality was implemented to clear the active session and return the user to the login interface.

7.5 Displaying Machine Risk Information

Presenting prediction results in a meaningful way was another challenge. Users need to understand machine health and risk information without dealing with complex technical details.

The interface was designed to display health scores, risk levels, and prediction information in a readable format.

7.6 Interface Styling

The default Streamlit appearance did not fully match the desired project design. Custom HTML and CSS were used to improve the appearance of headings, cards, navigation elements, buttons, and page sections

The styling was refined to improve readability and provide a consistent FactoryOps design.

7.7 Testing Different Modules

Testing each module separately and then checking the complete workflow required time and coordination.

The team tested the login flow, API connection, data display, navigation, refresh operation, logout operation, and error handling to improve the overall reliability of the application.

8. Learnings & Skills Acquired

During the internship, we gained practical knowledge of:

- Python programming
- Streamlit frontend development
- FastAPI backend development
- REST API integration
- HTML and CSS styling
- Machine and sensor monitoring
- Predictive maintenance concepts
- Application testing and debugging
- Teamwork and communication

9. Testimonials from team

The FactoryOps project provided practical exposure to developing an application for an industrial use case. The team gained experience in understanding requirements, designing application modules, connecting frontend and backend components, and testing a complete software workflow.

The project helped us understand how predictive maintenance concepts can be converted into a practical monitoring platform. Working on the dashboard, machine monitoring, sensor readings, prediction information, maintenance recommendations, and incident tracking modules improved our technical understanding.

The internship also helped us improve our teamwork, communication, time management, and problem-solving abilities. We learned that successful project development requires continuous testing, coordination, documentation, and improvement.

Overall, the project was a valuable learning experience because it connected software development concepts with a real-world industrial problem.

10. Conclusion

The FactoryOps project was developed as an AI-based industrial maintenance and process optimization system. The platform provides a centralized interface for machine monitoring, sensor tracking, prediction analysis, maintenance management, and incident monitoring.

The project combines a Streamlit frontend with a FastAPI backend to provide an organized and interactive application. The system supports users in reviewing machine conditions, understanding equipment risk, identifying possible maintenance requirements, and monitoring operational incidents.

During the eight-week internship, the team gained practical experience in Python, Streamlit, FastAPI, REST APIs, frontend design, backend integration, testing, and documentation. The project also improved our understanding of predictive maintenance and its importance in industrial environments.

FactoryOps demonstrates how software applications can support industrial organizations by improving machine visibility, organizing maintenance information, and assisting maintenance teams in making informed decisions.

The internship experience aligns with our academic and career goals by providing practical exposure to application development, system integration, problem-solving, teamwork, and real-world technology implementation.

11. Acknowledgements

We express our sincere gratitude to Infosys Springboard for providing us with the opportunity to participate in the Virtual Internship 7.0 program.

We would like to thank our mentor, Selciya mam , for her valuable guidance, encouragement, feedback, and continuous support throughout the internship.

We are also thankful to our team members for their cooperation and contributions during the development of the FactoryOps project. Finally, we thank everyone who supported us in successfully completing this internship.